

LEVEL 25 — Numerical Operations

Easy NumPy examples with line-by-line code and output.

1. Rounding — `np.round()`

Explanation: Rounds numbers to a selected number of decimal places.

Code:

```
import numpy as np

arr = np.array([1.234, 2.567, 3.891])
result = np.round(arr, 2)
print(result)
```

Output:

```
[1.23 2.57 3.89]
```

Important: 2 means keep 2 decimal places.

2. Trigonometric Functions

Explanation: NumPy provides `sin`, `cos`, and `tan`. Angles are in radians.

Code:

```
import numpy as np

angle = np.pi / 2

print(np.sin(angle))
print(np.cos(angle))
print(np.tan(0))
```

Output:

```
1.0
6.123233995736766e-17
0.0
```

Important: $\cos(\pi/2)$ is mathematically 0; the tiny value is floating-point precision.

3. Logarithmic Functions — `np.log()`

Explanation: `np.log()` calculates the natural logarithm.

Code:

```
import numpy as np

arr = np.array([1, np.e, np.e**2])
result = np.log(arr)
print(result)
```

Output:

```
[0.  1.  2.]
```

Important: $\ln(1)=0$, $\ln(e)=1$, and $\ln(e^2)=2$.

4. Exponential Functions — `np.exp()`

Explanation: `np.exp(x)` calculates e raised to the power x .

Code:

```
import numpy as np

arr = np.array([0, 1, 2])
result = np.exp(arr)
print(result)
```

Output:

```
[1.          2.71828183  7.3890561 ]
```

Important: $e^0=1$, $e^1\approx 2.718$, and $e^2\approx 7.389$.

5. Absolute Values — `np.abs()`

Explanation: Returns the distance of each number from zero.

Code:

```
import numpy as np

arr = np.array([-10, -5, 0, 5, 10])
result = np.abs(arr)
print(result)
```

Output:

```
[10  5  0  5 10]
```

Important: The absolute value of -10 is 10.

6. Sign Functions — `np.sign()`

Explanation: Shows whether each number is negative, zero, or positive.

Code:

```
import numpy as np

arr = np.array([-10, 0, 25])
result = np.sign(arr)
print(result)
```

Output:

```
[-1  0  1]
```

Important: -1 = negative, 0 = zero, 1 = positive.

7. Clipping

Explanation: Clipping keeps values inside a minimum and maximum limit.

Code:

```
import numpy as np

arr = np.array([5, 15, 25, 35])
result = np.clip(arr, 10, 30)
print(result)
```

Output:

```
[10 15 25 30]
```

Important: Values below 10 become 10; values above 30 become 30.

8. np.clip()

Explanation: np.clip(array, minimum, maximum) limits every value to the given range.

Code:

```
import numpy as np

marks = np.array([20, 45, 75, 110])
result = np.clip(marks, 0, 100)
print(result)
```

Output:

```
[ 20  45  75 100]
```

Important: 110 becomes 100 because the maximum is 100.

9. np.maximum()

Explanation: Compares two arrays element by element and returns the larger value.

Code:

```
import numpy as np

a = np.array([10, 50, 30])
b = np.array([20, 40, 35])
result = np.maximum(a, b)
print(result)
```

Output:

```
[20 50 35]
```

Important: Each position returns the larger number.

10. np.minimum()

Explanation: Compares two arrays element by element and returns the smaller value.

Code:

```
import numpy as np

a = np.array([10, 50, 30])
b = np.array([20, 40, 35])
result = np.minimum(a, b)
print(result)
```

Output:

```
[10 40 30]
```

Important: Each position returns the smaller number.

Quick Revision

- np.round() → rounding

- `np.sin()`, `np.cos()`, `np.tan()` → trigonometry
- `np.log()` → natural logarithm
- `np.exp()` → exponential
- `np.abs()` → absolute value
- `np.sign()` → sign of a number
- `np.clip()` → limits values to a range
- `np.maximum()` → larger value
- `np.minimum()` → smaller value